Extended reading — Week 9: Proximal Policy Optimization

November 8, 2026

1. Schulman, Wolski, Dhariwal, Radford, Klimov (2017) — Proximal Policy Optimization Algorithms

Preprint, arXiv:1707.06347.

Summary. PPO alternates interaction with surrogate optimization and enables repeated minibatch updates on a collected batch. The clipped and adaptive KL-penalty variants have favorable empirical comparisons on continuous-control and Atari tasks; these experiments are not convergence proofs.

Why it matters for this week. Derive the four cases formed by advantage sign and ratio location relative to the clipping interval. Separate improvement incentives from penalties for harmful changes.

Annotated key passage (abstract).

> *"a novel objective function that enables multiple epochs of minibatch updates"*

Batch reuse explains why old log probabilities stay fixed. The clipped surrogate is pointwise no larger than the unclipped surrogate, because it takes their minimum. This is not a lower bound on actual policy return.

2. Huang, Dossa, Raffin, Kanervisto, Wang (2022) — The 37 Implementation Details of Proximal Policy Optimization

ICLR Blog Track, 25 March 2022. Article and reference implementation.

Summary. The authors trace implementation revisions and reproduce results across classic control, Atari, robotics, recurrent policies, and multidiscrete actions. Their 37-item checklist contains task-specific choices; every item is not required for every environment.

Why it matters for this week. Audit stored Gaussian samples, action clipping, normalization, and rollout boundaries. The course's detail numbers are local labels, not an exact copy of the article's ordering.

Annotated key passage (introduction).

> *"conclusions are in MuJoCo tasks and do not necessarily transfer to other games such as Atari"*

Evaluate choices in their intended task. An ablation label is meaningful only if disabling the option changes the implementation.

3. Engstrom, Ilyas, Santurkar, Tsipras, Janoos, Rudolph, Madry (2020) — Implementation Matters in Deep Policy Gradients

ICLR 2020, arXiv:2005.12729.

Summary. Controlled PPO/TRPO comparisons show that code-level optimizations affect both measured performance and the trust-region behavior attributed to the algorithm. Comparisons can change when implementation conditions change.

Why it matters for this week. Match architectures, optimizers, interaction budgets, and evaluation protocols when comparing losses.

Annotated passage (paraphrase of the abstract).

> Implementation details affect the performance and behavior attributed to a policy-gradient algorithm.

This is a paraphrase, not a direct quotation. The result motivates controlled comparisons; it does not establish that PPO and TRPO are interchangeable on every task or that one checklist eliminates their differences.

4. Andrychowicz, Raichuk, Stanczyk, Orsini, Girgin, Marinier, Hussenot, Geist, Pietquin, Michalski, Gelly, Bachem (2021) — What Matters In On-Policy Reinforcement Learning? A Large-Scale Empirical Study

ICLR 2021, arXiv:2006.05990.

Summary. This broad empirical study investigates design decisions in on-policy deep RL, covering objectives, architectures, optimization, and data collection. The effects are conditional on the tested environments and protocols.

Why it matters for this week. Select an ablation from the study, then state its task, budget, and seeds before interpreting the effect.

Annotated passage (paraphrase of the abstract).

> Empirical investigation of many design choices can inform on-policy RL implementations.

This is a paraphrase, not a direct quotation. Recommendations motivate measurements; they are not convergence theorems. There is no universal ordering of which implementation choice matters most.

5. Ouyang, Wu, Jiang, et al. (2022) — Training language models to follow instructions with human feedback (InstructGPT)

NeurIPS 2022, arXiv:2203.02155.

Summary. InstructGPT combines supervised demonstrations, a preference reward model, and PPO fine-tuning to improve instruction following under human evaluation. It is an application of PPO, not evidence that all preference-training systems use the same optimizer.

Why it matters for this week. Distinguish the old rollout policy from the SFT reference used in KL reward regularization.

Annotated passage (paraphrase of the method).

> Divergence from the supervised reference is penalized during reward optimization.

This is a paraphrase, not a direct quotation. A finite penalty discourages drift but does not eliminate reward hacking or guarantee safety. Week 18 supplies a small synthetic analogue, rather than a frontier-model reproduction.

Sourced from

`research/deeprl_core_sota.md` §2.6 and `research/llm_rl_sota.md` (RLHF section).

Statistics reading bridge

A positive four-seed average need not resolve a PPO comparison. Read mean intervals and paired differences before interpreting ablation winners. State what was matched, how checkpoints were selected and which evaluation outcomes remained untouched; use PPO sources for clipping behavior.

Checked textbook locations: Siegrist § 8.2, PDF p.966; Ross § 7.3, PDF p.309. https://stats.libretexts.org/@go/page/10201. These locations support the probability and estimation concepts; the algorithm-specific results remain attributed to the primary readings above. The course source catalog distinguishes the original textbook from its supplied instructor manual and records PDF-viewer pagination.